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ORIGINAL ARTICLE

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Catalytic hydrothermal depolymerization of nylon 6

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Abstract Depolymerization of nylon 6 to produce ε-caprolactam using an environmentally friendly heteropoly acid catalyst was studied at temperatures between 553 and 603 K in water. The products of depolymerization were analyzed qualitatively and quantitatively by means of mass spectrometry and high-performance liquid chromatography. The results showed that the depolymerized product was mainly ε-caprolactam with a little 6-aminocaproic acid and oligomers. The phosphotungstic heteropoly acid used as a catalyst can improve the hydrolysis rate and yield of ε-caprolactam. The optimum hydrolysis conditions for ε-caprolactam yield were as follows: phosphotungstic heteropoly acid content, 3%; reaction temperature, 573 K; and reaction time, 85 min. Under these conditions, the yield of ε-caprolactam was 77.96%. In the temperature range 553–603 K, the activation energy of 3% phosphotungstic heteropoly acid-catalyzed depolymerization was evaluated as 77.38 kJ/mol, which is lower than the 86.64 kJ/mol value for no catalyst.

Key words Nylon 6 · Phosphotungstic heteropoly acid · Depolymerization · ε-Caprolactam · Kinetics

Introduction

Plastic products are widely used in human life, and waste plastics can cause serious pollution. The use of pyrolysis to convert waste plastic into chemical intermediates is an important process, but up to now, the most common application of direct pyrolysis in an inert atmosphere has had many disadvantages such as batch production, low yield, poor-quality products, and long reaction time; in addition, secondary pollution can occur through dust, exhaust air, wastewater, and solid waste and high levels of carbonization can block tubing. Thus, the current industrial application of pyrolysis is very discommodious. Recently, the application of high-temperature and high-pressure water to pyrolysis has received much attention for its unique properties such as dissolution of nonpolar organic substances, its low dielectric properties and high diffusibility, and its high concentrations of H^+ and OH^- ions from the dissociation of water.[1–5] Pyrolysis of waste plastic in high-temperature and high-pressure water can avoid the disadvantages of direct pyrolysis and shows good prospects for recycling waste polymers to produce useful chemicals.[6]

Nylon 6 (PA 6) is an important engineering plastic and is widely used; consequently, development of an efficient treatment system for waste PA 6 is increasingly urgent. As a polymer with acyl and amido linkages, it was found that PA 6 can hydrolyze at acyl–amido bonds to ultimately produce ε-caprolactam in high-temperature and high-pressure water.[7–9] With increasing temperature, the hydrolysis rate of PA 6 increases, but the ε-caprolactam product decomposes quickly. A suitable catalyst could play an important role by increasing the hydrolysis rate and attaining a high yield of ε-caprolactam at comparatively low temperatures.[10] Therefore, we adopted phosphotungstic heteropoly acid ($H_3PW_{12}O_{40}$, HPA) as a catalyst to accelerate hydrolysis of PA 6 and increase the yield of ε-caprolactam in comparatively mild hydrothermal conditions. HPA as an acid catalyst presents strong acidity, high hydrothermal stability, and low acid corrosion to most metals owing to the formation of passivation layers.[11]

Experimental

The hydrothermal vessel used had a volume of 18 ml and was heated in an electric salt bath of 1:1 (Wt/Wt) $NaNO_3$

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and KNO_3. The ratio of nylon:water was about 1:15 (Wt/Wt) and the total volume of all reactants placed in the vessel was 12 ml. After the reaction, the hydrothermal vessel was put in ice water to cool it to room temperature and was then opened and the product was removed for analysis.

Liquid-phase products after HPA catalysis were neutralized by 0.1 mol/l NaOH solution before measurement and were analyzed by mass spectrometry (MS) and high-performance liquid chromatography (HPLC). The MS instrument used was a Hybrid Quadrupole-TOF system, Applied Biosystems, USA. The measurements used an electron spray ionization (ESI) of 4.5 kV and 573 K, a GS1 of 45.00, a GS2 of 30.00, a CUR of 10.00 μA, and an m/z range of 90–1000. The HPLC system used was an Agilent 1100 system, Agilent, USA. The measurements used a Zorbax SB-C18 chromatographic column (5 μm, 250 mm × 4.6 mm) with a UV-Vis detector at 225 nm, a column temperature of 303 K, a mobile phase of water: acetonitrile ratio of 80:20±0.1% (V/V) acetic acid, a flow velocity of 0.1 ml/min, and a sample size of 5 μl.

Results and discussion

Figure 1 shows the temperature history within the reactor during a run. According to the figure, it took about 2 min to reach the desired temperature. Therefore, the starting time for a reaction was defined as 2 min after the loading of the reactor into the salt bath.

Figure 2 shows a typical MS spectrum of the liquid products; the compounds corresponding to the main peaks are given in Table 1. The results clearly show that the main product is ε-caprolactam; minor products include 6-aminocaproic acid and its dimer, trimer, tetramer, and pentamer and oligomers of ε-caprolactam. These results indicate that PA 6 is depolymerized to ε-aminocaproic acid followed by cyclodehydration to ε-caprolactam (according to a previous study) through hydrolysis, i.e., PA6 → ε – aminocaproic acid → ε – caprolactam.[9]

To help determine the optimum conditions for PA 6 hydrolysis to produce ε-caprolactam, the depolymerization rate of PA 6 is expressed as $(M_0–M_t)/M_0 \times 100\%$, where M_0 and M_t are the weight of initial PA 6 and residual solid after the reaction, respectively. The yield of ε-caprolactam is expressed as $m_t/M_0 \times 100\%$, where m_t is the weight of ε-caprolactam produced.

The depolymerization rate of PA 6 and the yield of ε-caprolactam when 3% HPA was used as a catalyst and when no catalyst was used at 553 K are shown in Figs. 3 and 4. With no catalyst, there is practically no depolymerization of PA 6 at 25 min, and near total depolymerization is attained at about 115 min; the yield of ε-caprolactam at this time is only 65.62%. With 3% HPA used as catalyst, the time for complete depolymerization is about 100 min, and at 115 min, the yield of ε-caprolactam is 70.54%. These results show that HPA can accelerate depolymerization of PA 6 and increase the yield of ε-caprolactam.

Figures 5 and 6 give the depolymerization rate of PA 6 and the yield of ε-caprolactam at different temperatures. With increasing temperature, the time of total depolymerization of PA 6 decreases. At 603 K, it is only 25 min, whereas the times of total depolymerization at 553, 573, and 583 K are 100, 70, and 55 min, respectively. These results show that high temperature benefits the depolymerization of PA 6.

Table 1. Compounds corresponding to the peaks in the mass spectrum shown in Fig. 2

m/z	Compound
114.1	ε-Caprolactam
132.1	6-Aminocaproic acid
227.2	ε-Caprolactam dimer
245.2	6-Aminocaproic acid dimer
340.2	ε-Caprolactam trimer
358.3	6-Aminocaproic acid trimer
471.4	6-Aminocaproic acid tetramer
584.5	6-Aminocaproic acid pentamer

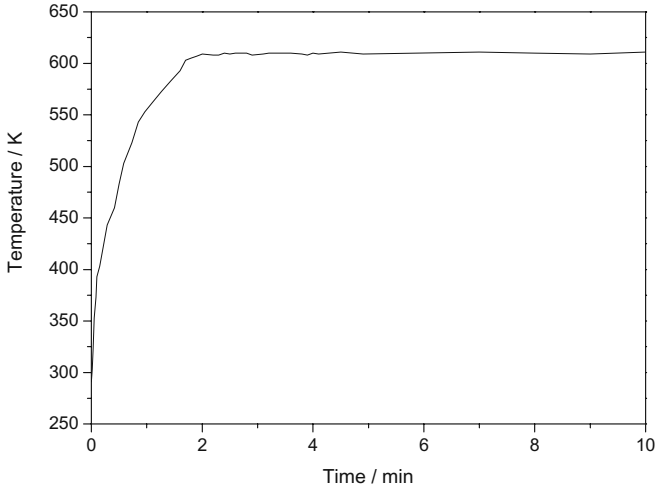

Fig. 1. Temperature change in the reactor as a function of time (set temperature: 610 K)

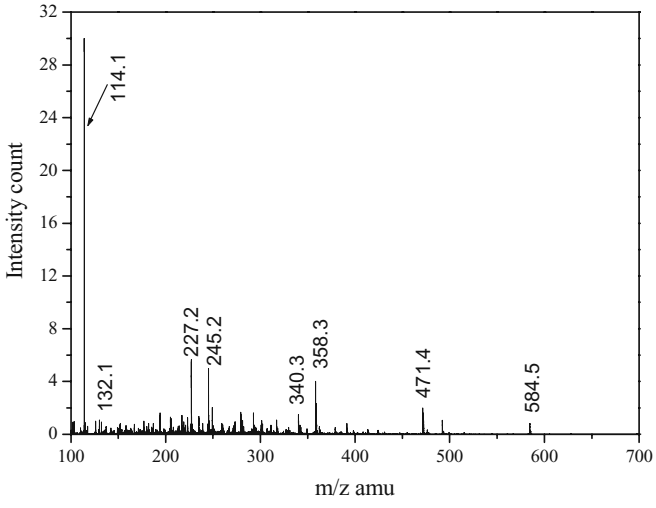

Fig. 2. Typical mass spectrum of liquid products [593 K, 40 min, and 3% phosphotungstic heteropoly acid (HPA)]

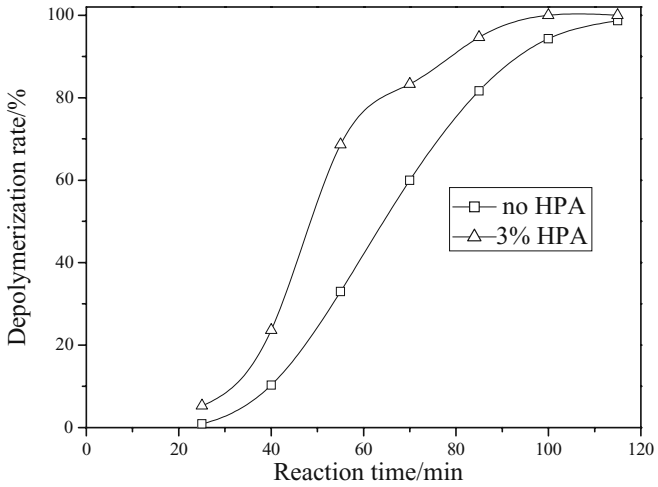

Fig. 3. Catalytic and noncatalytic depolymerization rates at 553 K

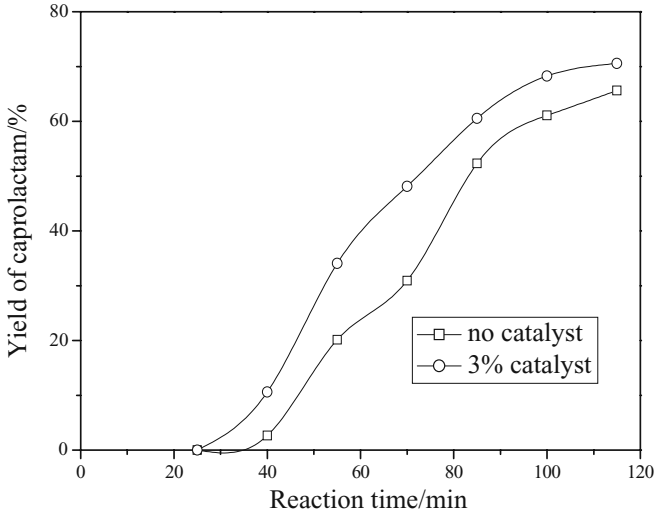

Fig. 4. Catalytic and noncatalytic ε-caprolactam yields at 553 K

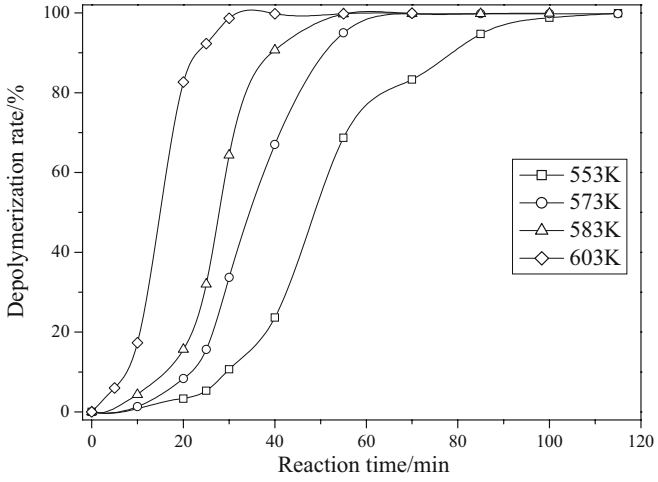

Fig. 5. Depolymerization rate of PA 6 with reaction time at different temperatures (3% HPA)

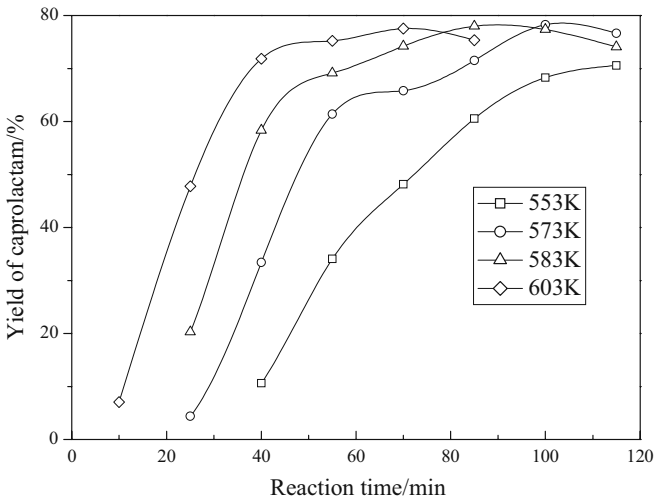

Fig. 6. Yield of ε-caprolactam with reaction time at different temperatures (3% HPA)

At comparatively short times, the yield of ε-caprolactam shows similar behavior to that of depolymerization, i.e., it increases with increasing time; however, the yield of ε-caprolactam shows transitional behavior at comparatively long times. For example, for the highest temperature examined, 603 K, the highest yield of ε-caprolactam is 77.52% at 70 min, but the yield decreases to 75.32% at 85 min. This phenomenon is a result of ε-caprolactam decomposition with time at high temperature.[9] Therefore, for hydrothermal depolymerization of PA 6 to produce ε-caprolactam, temperature and time are both important factors affecting the yield; thus, determination of the appropriate time and temperature is very important. According to the relation of the yield of ε-caprolactam with temperature and time, the optimum conditions when using 3% HPA as catalyst are a temperature of 583 K and a time of 85 min; under these conditions, ε-caprolactam can attain the highest yield of 77.96%.

Polymer depolymerization can be treated as pseudo-first-order reaction. Thus, the equation of PA 6 depolymerization kinetics can be expressed as follows:

$$-d[\text{PA 6}]/dt = k[\text{PA 6}][H_2O]^n \quad (1)$$

Owing to an excess of water in the reaction mixture, Eq. 1 can be simplified as:

$$-d[\text{PA 6}]/dt = k'[\text{PA 6}] \quad (2)$$

If the initial content of PA 6 is expressed as $[\text{PA 6}]_0$, and the content at time t is expressed as $[\text{PA 6}]_t$, then the integral of Eq. 2 can be expressed as:

$$\ln[\text{PA 6}]_0/[\text{PA 6}]_t = k't \quad (3)$$

Because PA 6 is solid, its content can be expressed in terms of weight. Therefore, the expression $[\text{PA 6}]_0/[\text{PA 6}]_t$ can be replaced with M_0/M_t, where M_0 and M_t are the initial weight and the weight at time t, respectively. Thus Eq. 3 can be expressed as $\ln(M_0/M_t) = k't$. Table 2 gives degradation rate constant k′ at different temperatures.

The data for constant k′ and temperature are linearly fitted as shown in Fig. 7. According to the Arrhenius equation, the linear relation of lnk′ and 1000/T gives the reaction activation energies as 86.64 and 77.38 kJ/mol for no catalyst and 3% HPA, respectively. Pre-exponential factors A are 4.31×10^6 and 1.05×10^6 for no catalyst and 3% HPA, respectively. It has been reported that if the decomposition of PA 6 is described by the following equation, then the apparent activiation energy (Ea_1) is 97.1 kJ/mol at 573–633 K.[9]

$$\text{PA 6} \xrightarrow{k_1} \varepsilon - \text{aminocaproic acid} \quad (4)$$

Table 2. Constant of catalytic depolymerization (k′) at different temperatures

Temperature (K)	553	573	583	603
k′ (min^{-1}) (3% HPA)	0.0485	0.0965	0.1142	0.2003
k′ (min^{-1}) (no HPA)	0.0485	0.0965	0.1142	0.2003

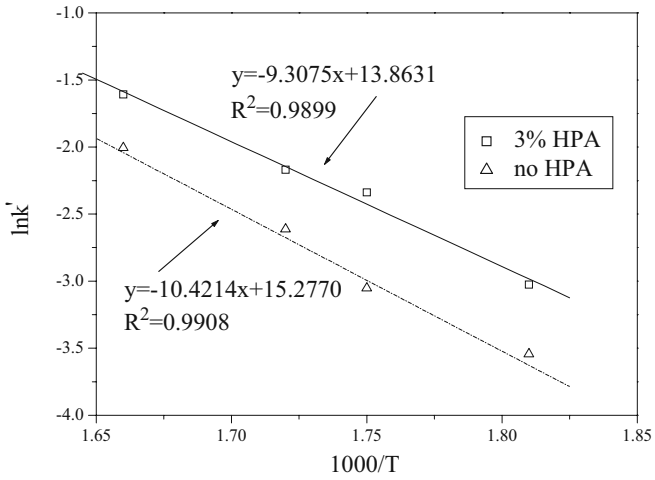

Fig. 7. Relation of rate constants (*k′* (3% HPA) and (no HPA)) to temperature

Because Ea_1 was obtained using the concentration of the products, the value is higher than using the concentration of [PA 6], which indicates that although the residue solid cannot depolymerize to ε-aminocaproic acid and then ε-caprolactam, it does consist of depolymerization product. Figure 8 gives the thermogravimetry (TG) and differential scanning calorimetry (DSC) curves of PA 6 and solid residue at 603 K and 20 min. The TG curves show that the pyrolysis temperature of the residue is clearly lower than that of the sample of PA 6. The residue starts to decrease in weight at a temperature of about 250°C and attains maximal decrease of weight at about 431°C; the corresponding temperatures are about 356°C and 464°C for PA 6. The DSC curve shows that the melting temperature of the residue is about 196°C, which is lower than the corresponding temperature of 221°C for the PA 6 sample. These results indicate that the thermodynamic stability of the residue is lower than that of the original PA 6 sample and the residue should be a lower molecular weight polymer than the sample, thus, the residue is likely to be the depolymerization product of the sample.

To achieve separation of ε-caprolactam from the HPA solution, the new method of pervaporation through cross-linked poly vinyl alchol (PVA) membranes was adopted.[12,13] Figure 9 gives ultraviolet–visible absorption spectra of 3% HPA solution and residual HPA solution after pervaporation of ε-caprolactam. The peak at about 265 nm results from $PW_{12}O_{40}^{3-}$ dissociation from HPA. This result indicates that HPA is stable after being used, and thus the residual HPA solution can be recycled.

Conclusion

PA 6 hydrolyzes in subcritical water to yield a main product of ε-caprolactam. This method is very important not only for treatment of waste PA 6 but also for recovery of ε-caprolactam, a useful resource. The main ε-caprolactam compounds decompose quickly in high-temperature supercritical water, thus lowering the yield. Addition of an appropriate amount of HPA can improve PA 6 hydrolysis to give

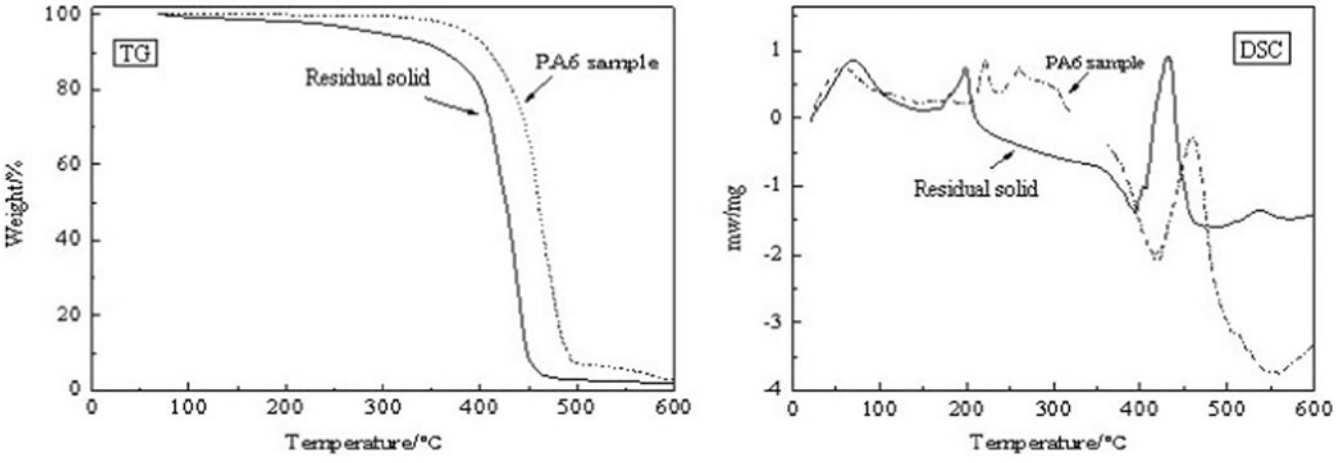

Fig. 8. Thermogravimetry (*TG*) and differential scanning calorimetry (*DSC*) curves of nylon 6 (*PA 6*) and the solid residue at 603 K and 20 min

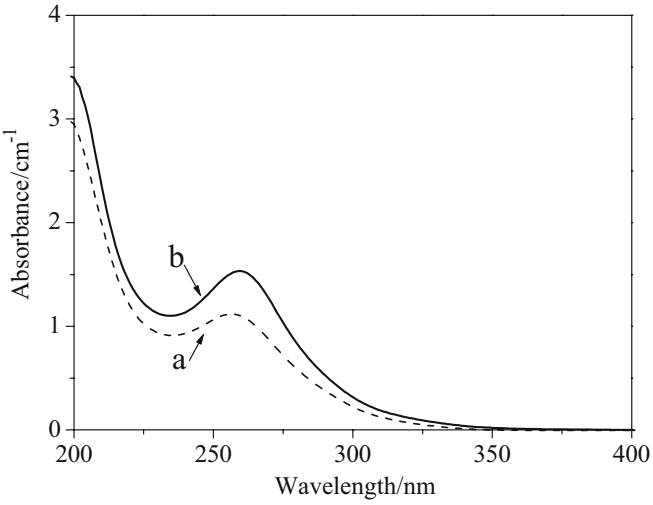

Fig. 9. Ultraviolet–visible spectrophotometry absorption spectra of HPA solution: *a*, residual HPA solution after pervaporation of ε-caprolactam at 3% HPA, 573 K, and 85 min; *b*, 3% HPA solution

high yields of ε-caprolactam at comparatively mild hydrothermal conditions, and the decrease of apparent activation energy with 3% HPA indicates that the addition of HPA can indeed accelerate the depolymerization of PA 6.

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